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AI-Based Phishing URL Detection System

ABSTRACT

The AI-Based Phishing URL Detection System is an intelligent application designed to help users identify malicious web links before they interact with them. The main objective of this project is to develop a user-friendly system that analyses a given URL, classifies it as legitimate or phishing, and presents the result along with a confidence score to support safer web browsing.

The proposed system uses Artificial Intelligence and Machine Learning techniques to analyse the structure and characteristics of URLs. Python, Pandas, and scikit-learn are used to prepare the data and train the classification models. From each URL, the system extracts features such as URL length, presence of special symbols like "@" and "-", use of an IP address in place of a domain name, number of subdomains, HTTPS usage, suspicious keywords, and the use of URL shortening services. Based on these features, machine learning classifiers such as Logistic Regression, Support Vector Machine, Random Forest, and XGBoost are trained and compared using accuracy, precision, recall, and F1-score, and the best-performing model is selected for detection.

The application also provides an interactive web interface in which users can enter or paste a URL, check it instantly, and view the prediction result in real time. Because the model learns patterns from labelled data rather than depending only on blacklists, it can identify new and previously unseen phishing URLs. The system can be retrained with fresh data, making it easy to keep up with changing phishing techniques. It aims to make phishing awareness and URL checking more accessible and convenient for everyday internet users.

This project demonstrates how AI and machine learning can be applied to cybersecurity and online fraud prevention. The system is intended as a supplementary security aid and does not replace antivirus software, secure browsing practices, or the guidance of cybersecurity professionals.

Keywords: Artificial Intelligence, Phishing Detection, Machine Learning, URL Analysis, Feature Extraction, Random Forest, Cybersecurity, Web Security.

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